

Assignment – 10:

Problem Statement:

Write a program in C to simulate a simple calculator that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The calculator should also perform special operations like computing x^y (power) and $x!$ (factorial).

Code:

```
#include <stdio.h>

int factorial(int n)
{
    int fact=1;
    for (int i=1;i<=n;i++)
    {
        fact=fact*i;
    }
    return fact;
}

int main()
{
    int choice,n;
    float a,b,result;
    printf("Simple Calculator\n");
    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("5. Power (x^y)\n");
    printf("6. Factorial (x!)\n");
    printf("Enter your choice: ");
```

```
scanf("%d",&choice);
switch (choice)
{
    case 1: printf("Enter two numbers: ");
            scanf("%f%f",&a,&b);
            printf("Sum = %.2f\n",a+b);
            break;
    case 2: printf("Enter two numbers: ");
            scanf("%f%f",&a,&b);
            printf("Difference = %.2f\n",a-b);
            break;
    case 3: printf("Enter two numbers: ");
            scanf("%f%f",&a,&b);
            printf("Product = %.2f\n",a*b);
            break;
    case 4: printf("Enter two numbers: ");
            scanf("%f%f",&a,&b);
            if (b!=0)
            {
                printf("Result = %.2f\n",a/b);
            }
            else
            {
                printf("Error! Division by zero.\n");
            }
            break;
    case 5: printf("Enter base and exponent: ");
            scanf("%f%f",&a,&b);
            result=1;
```

```

        for (int i=1;i<=(int)b;i++)
        {
            result=result*a;
        }
        printf("Result = %.2f\n",result);
        break;
case 6: printf("Enter a number: ");
        scanf("%d",&n);
        if (n>=0)
        {
            printf("Factorial = %d\n",factorial(n));
        }
        else
        {
            printf("Factorial is not defined for negative numbers.\n");
        }
        break;
default: printf("Invalid Choice!\n");

    }
}

```

Input & Output:

```

Simple Calculator
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial (x!)
Enter your choice: 1
Enter two numbers: 23 42
Sum = 65.00

```

```

Simple Calculator
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial (x!)
Enter your choice: 2
Enter two numbers: 3 6
Difference = -3.00

```

Simple Calculator

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial ($x!$)

Enter your choice: 3

Enter two numbers: 56 2

Product = 112.00

Simple Calculator

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial ($x!$)

Enter your choice: 4

Enter two numbers: 94 2

Result = 47.00

Simple Calculator

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial ($x!$)

Enter your choice: 5

Enter base and exponent: 6 2

Result = 36.00

Simple Calculator

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial ($x!$)

Enter your choice: 6

Enter a number: 5

Factorial = 120

